Sheet 7

Choi matrices and dilations

1. For the decay channel $K_0 = \mathrm{diag}(\sqrt{1-\gamma}, 1)$, $K_1 = \sqrt{\gamma}\,|1\rangle\langle 0|$, $0 \le \gamma \le 1$, compute the unnormalised Choi matrix in the input-reference/output order $00, 01, 10, 11$. Find its eigenvalues and check its trace and output partial trace.
2. At $\gamma = 3/4$, write the isometry J and a complete unitary matrix on system plus a ready qubit that implements it. Use basis $00, 01, 10, 11$ and ready environment $|0\rangle$. Check orthonormal columns.
3. Compare the two Bell preparations $\Phi_\pm = (|00\rangle \pm |11\rangle)/\sqrt{2}$. Find their marginals, then apply the CNOT with system controlling environment. Compute the final system marginals. What information, absent from either initial marginal, determines their different futures?